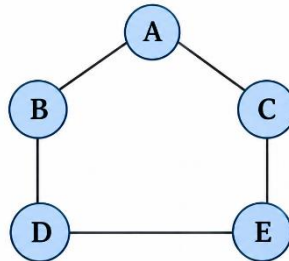


- Q1)** Write a program to solve a state-space search problem using BFS. **[10 marks]**

Start Vertex = A



- Q2)** Write a program to implement Predicate Logic using predicates, variables, and quantifier. **[20 marks]**

OR

- Q2)** Write a program to implement Gaussian NB classifier with 3 classes on numeric data, use `make_blobs` to create a dataset X of 200 sample points in 2D Space, 3 clusters and each cluster spread (SD) over 1.5. For testing (Prediction), use the following values to test the Gaussian NB Classifier.
Input (Values) : [-2, 5], [0,0], [6, -0.3]
Output : [0 1 1]
Print the accuracy score using the `accuracy_score()` from `sklearn.metrics` library.. **[20 marks]**

- Q3)** Viva **[05 marks]**